



Response rate to FAO questionnaires

Jose Rosero Moncayo
Director, Statistics Division
Food and Agriculture Organization of the UN
Jose.RoseroMoncayo@fao.org



Measuring data quality

- FAO produces international statistics
- In order to do this it needs to compile, harmonize and standardize the information provided by countries
- Countries are our main source of data and as such FAO is heavily depended on the quantity and the quality of the data countries provide to us.
- Therefore, it is important to measure the level of response of the questionnaires by countries, and the level of completeness of the resulting databases with data coming from official sources.



Measuring data quality

- The Statistics Division of FAO (ESS) dispatches 6 annual questionnaires, and collects data for the following agricultural domains:
 - Agricultural Production
 - Producer prices
 - Fertilizers use
 - Pesticides use
 - Government Expenditures in Agriculture (GEA)
 - Land Use



Measuring data quality (ct'd)

- The Fisheries Division dispatches mainly 6 annual questionnaires, and collects data for the following agricultural domains:
 - Aquaculture Production
 - Capture production
 - Disposition of fishery production
 - Fleet
 - Number of fishers and fish farmers
 - Production of processed commodities



Measuring data quality

- Response rate indicator is defined here by:

$$\textit{Response rate} = \frac{\textit{Number of questionnaires filled (by year)}}{\textit{Number of questionnaires dispatched (by year)}}$$



Measuring data quality

- Data completeness indicator is defined here by:

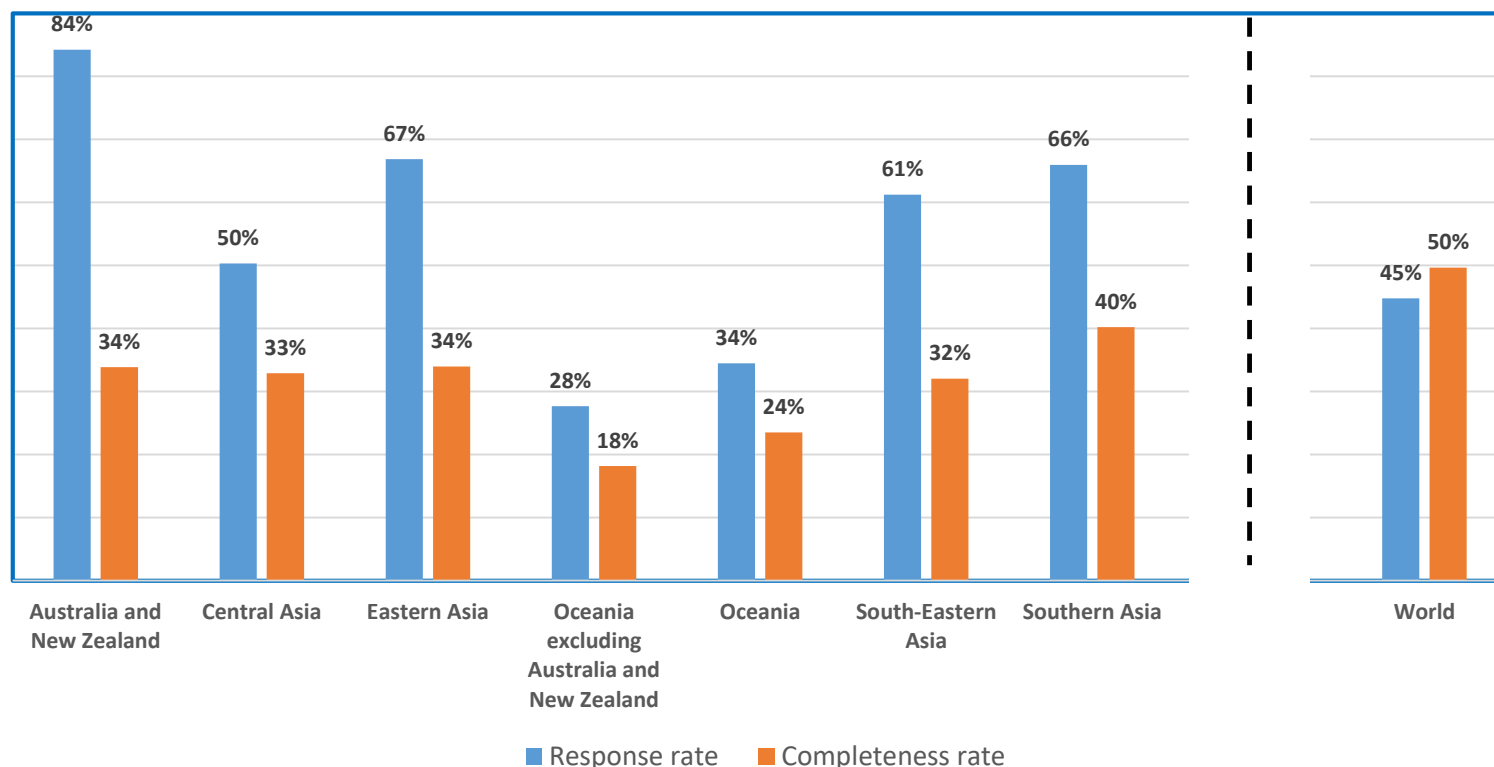
$$\text{Completeness rate} = \frac{\text{Number of official data points (by year)}}{\text{Number of expected data points (by year)}}$$

- The presentation will focus on the response indicator



Findings: The majority of the regions (except Pacific Islands) have a response rate that is above the average of the world (ESS-Questionnaires).

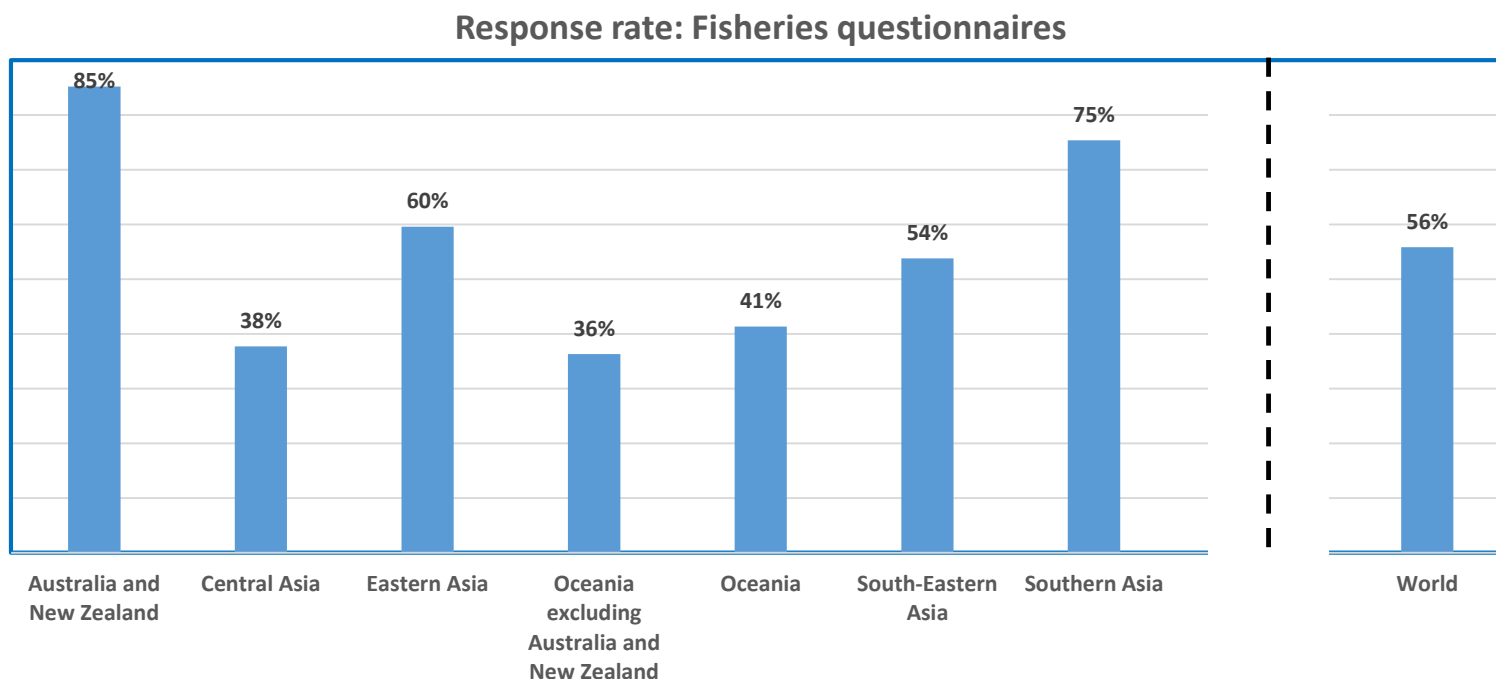
Average 2013-2018 response and completeness rates, by region





Findings: response rate for APCAS members is slightly below the world's average (Fisheries-Questionnaires).

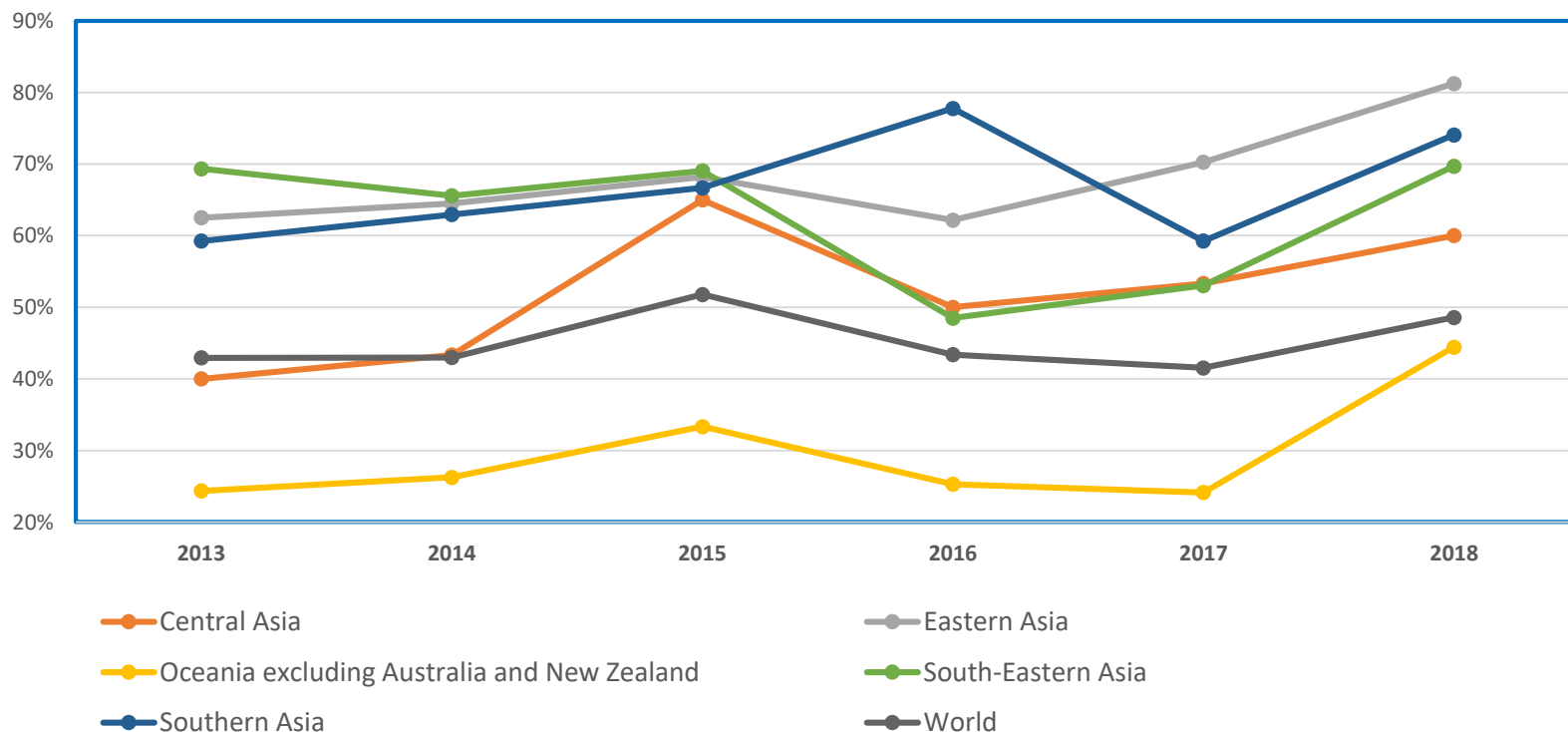
Average 2013-2018 response rates, by region





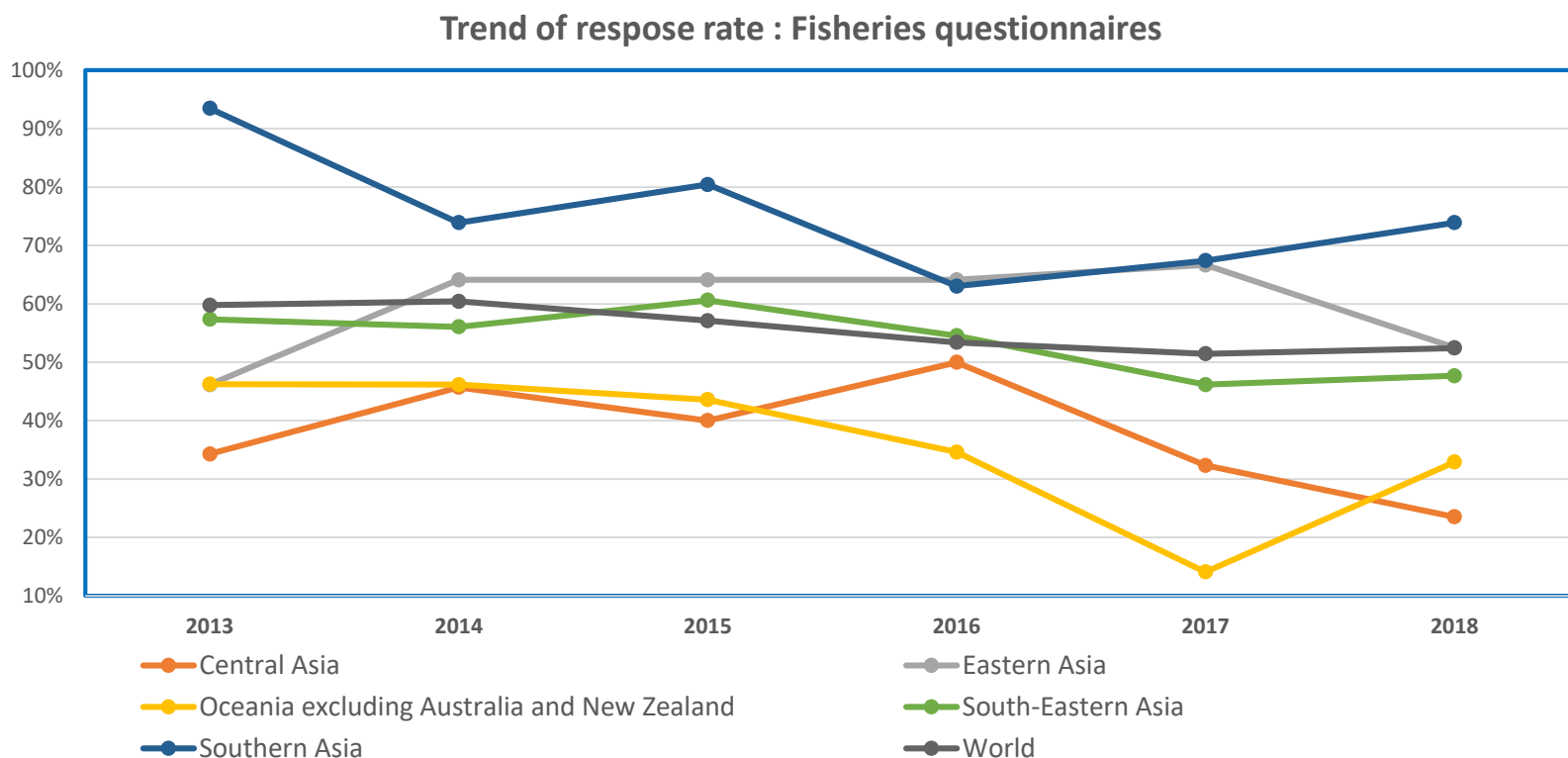
Findings: Across time there is general improvement in response rate for the ESS-AG Questionnaires

Trend of response rate: ESS Questionnaires





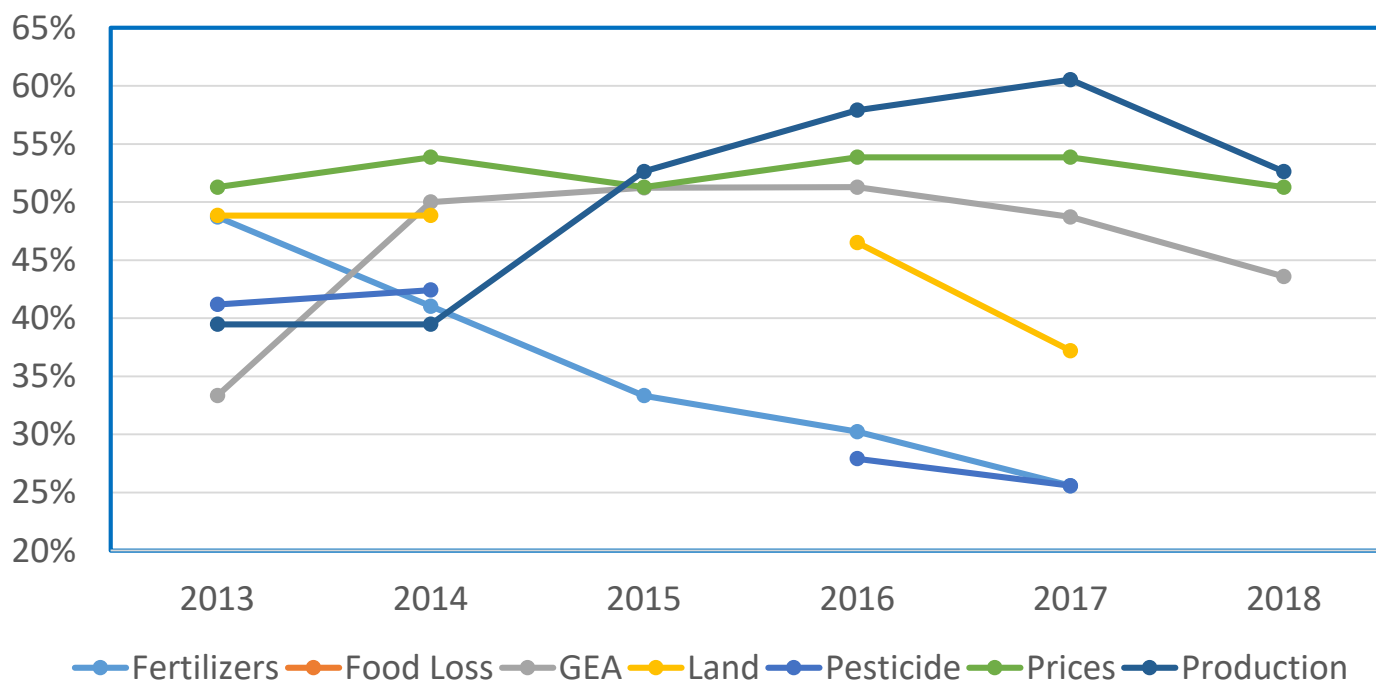
Findings: Across time and regions, there is a decrease in response rate for the Fisheries Questionnaires





Findings: Across domains, Prices and Production had higher response rates than other domains over time (ESS-Questionnaires)

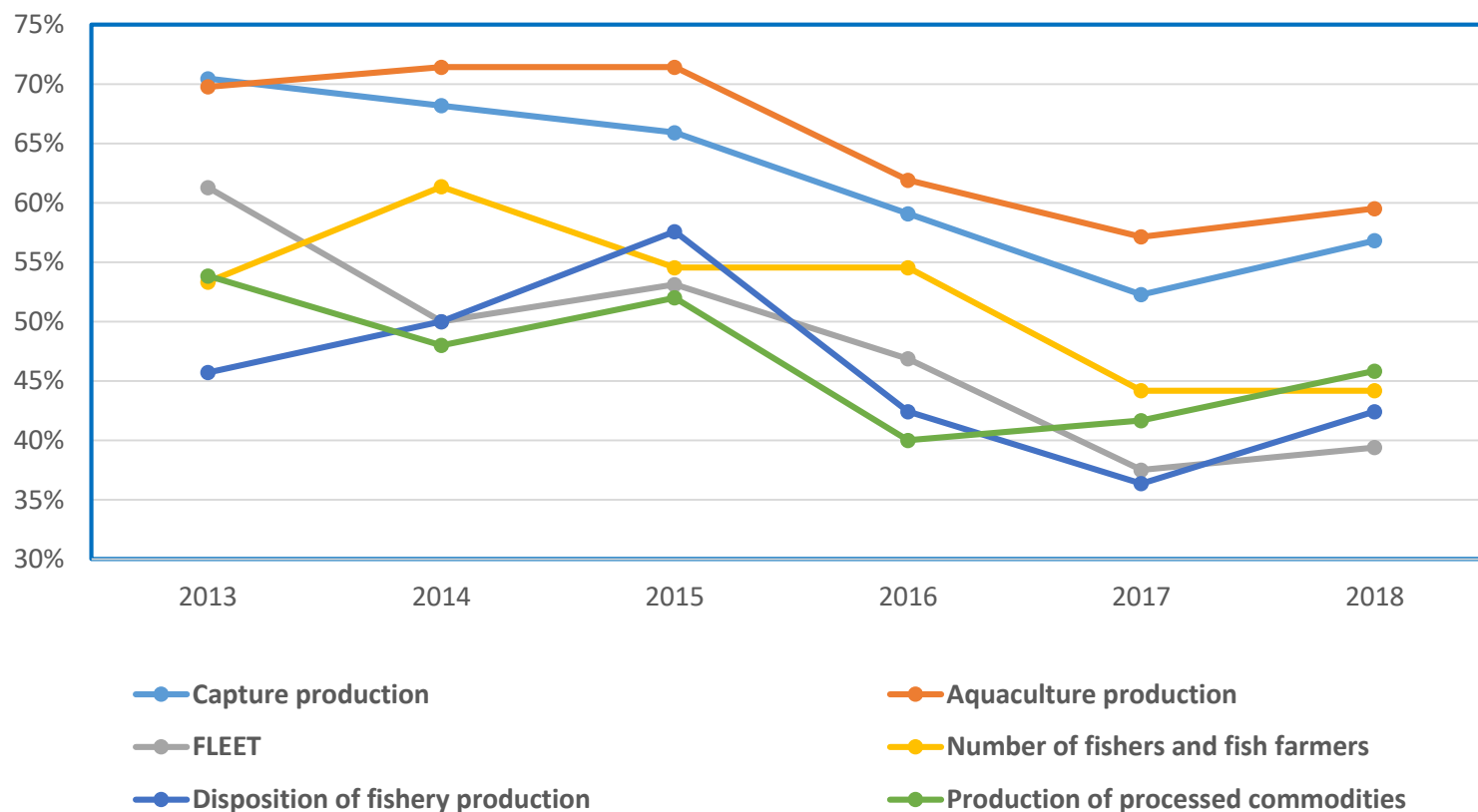
Trend of response rate by questionnaire





Findings: Aquaculture and Capture had higher response rates than other domains over time (Fisheries-Questionnaires)

Response rate APCAS: Fisheries Questionnaires





Why do we have a low or no response:

- Is it that we do not reach the countries
 - Lack of accuracy on list of relevant contact points at country level
- Is it that countries can not report due to different reasons like:
 - Increased burden: from FAO questionnaires (length, complexity, number, etc.); from all sources (regional/international organizations, NGOs, etc.)
 - Resource limitations: lack of expertise; budget/HR constraints at country level; staff turnover/loss of experience
 - Lack of data availability: less than annual frequency, or no data collection, in some domains (though this absence should be reported)



What can we do about it?: for discussion

- To improve in the coordination mechanisms:
 - Maintain up-to-date national focal point(s) with the support of the FAO regional and country offices
 - Continue to harmonize the questionnaire dispatch
- Develop alternative data reporting/ collection means
 - FAO pilots multiple-mode data collection, such as on-line questionnaires;
- Promote an open and continuous communication with the national focal point to ask for missing information or clarify doubts



What can we do about it?: for discussion

- FAO and member countries explore how best to implement statistical capacity development activities in the areas of data collection and data reporting/compilation
- Expand national data peer-review process for imputed data
- Others?



Food and Agriculture
Organization of the
United Nations

Thank you!

Jose.RoseroMoncayo@fao.org

28th Session of Asia and Pacific Commission on Agricultural Statistics
Bali, Indonesia, 10 – 14 February 2020